A Clinician's Guide to Common and Complex Dermatological Presentations: Management Plans for the AMC CAT Examination and Australian Clinical Practice

Introduction

This report provides an expert-level, evidence-based management plan for ten key dermatological conditions, specifically tailored to the requirements of the Australian Medical Council (AMC) Clinical Assessment Test (CAT) and the realities of clinical practice in Australia. The focus is on diagnostic precision, nuanced management strategies, and practical patient counseling to facilitate mastery of these topics. The conditions covered include common viral exanthems of childhood, infestations, bacterial infections, and more complex congenital and inflammatory dermatoses. The provided user document containing practice questions did not include specific scenarios for most of these dermatological conditions; however, where relevant (scabies, molluscum contagiosum, erysipelas), insights have been integrated. Therefore, this report is primarily synthesized from leading Australian clinical resources, including The Royal Children's Hospital (RCH) Melbourne Clinical Practice Guidelines, Therapeutic Guidelines, The Royal Australian College of General Practitioners (RACGP), and state health department protocols, ensuring maximum relevance and accuracy for the Australian context.

Section 1: Viral Exanthems of Childhood

The viral exanthems of childhood represent a spectrum of host-virus interactions. Some conditions, such as Hand, Foot, and Mouth Disease, are a direct consequence of viral replication, with the rash coinciding with systemic symptoms. In contrast, others like Roseola Infantum are characterized by a rash that appears as the viraemia resolves and fever subsides, reflecting an immune-mediated process. A third category, exemplified by Gianotti-Crosti syndrome, represents a more complex, delayed-type hypersensitivity reaction to a preceding infection. Understanding these distinct temporal patterns and underlying mechanisms is fundamental to moving beyond rote memorization to a deeper conceptual grasp, which is critical for accurate diagnosis and management in clinical practice.

Cocksackie Virus A16 (Hand, Foot, and Mouth Disease - HFMD)

- **Pathophysiology and Epidemiology**
  - HFMD is a common and highly contagious illness caused by enteroviruses, most frequently Coxsackievirus A16 and Enterovirus 71.
  - Transmission occurs via respiratory droplets from coughs and sneezes, direct

    - contact with fluid from blisters, and the faecal-oral route. The virus can be shed in faeces for several weeks after the illness has resolved.
    - It predominantly affects children under 10 years of age, with outbreaks being common in childcare centres and kindergartens due to the ease of transmission.
- **Clinical Presentation and Key Diagnostic Features**
    - **Prodrome:** The illness typically begins with a 1-2 day prodrome of low-grade fever, tiredness, a mild sore throat, and sometimes a runny nose.
    - **Enanthem (Oral Lesions):** This is followed by the appearance of painful small blisters and ulcers inside the mouth, on the tongue, gums, and inside the cheeks. This can make eating and drinking very uncomfortable for the child.
    - **Exanthem (Skin Rash):** A characteristic non-pruritic rash develops, consisting of small red spots that may progress to small blisters (vesicles). These are classically found on the palms of the hands and soles of the feet. The rash may also appear on the buttocks, and in infants, in the nappy area. The vesicles are often described as grey and elliptical in shape.
    - **Diagnosis:** The diagnosis of HFMD is clinical, based on the characteristic history and the typical distribution of the oral lesions and skin rash. Laboratory investigations are not routinely required.
- **Comprehensive Management Plan**
    - **Supportive Care:** Management is entirely symptomatic, as there is no specific antiviral treatment for the causative viruses. Antibiotics are not effective and should not be prescribed.
    - **Hydration:** Maintaining adequate hydration is the primary goal. Due to painful mouth ulcers, children may refuse to drink. Parents should be advised to offer frequent sips of water or oral rehydration solutions. Cold items like icy poles or yoghurt can be soothing and provide fluids. For younger children, administering fluids via a syringe may be necessary.
    - **Analgesia:** Simple analgesia, such as paracetamol or ibuprofen, is crucial for managing the pain from mouth ulcers and controlling fever. This should be given regularly to improve comfort and encourage oral intake. It is imperative to advise parents to avoid giving aspirin to children due to the risk of Reye's syndrome.
    - **Skin Care:** Blisters on the hands and feet should be left to dry naturally. They should not be pierced or squeezed, as the fluid within them is infectious and can spread the virus.
- **Referral Pathways and Red Flags**
    - A critical aspect of assessing any child with a fever and rash is to exclude more serious conditions, particularly meningococcal disease. The presence of a non-blanching rash (petechial or purpuric) is a medical emergency. The "press test" (pressing a clear glass against the rash) is a vital screening tool; if the spots do not fade, the child requires immediate transfer to a hospital.
    - **Urgent Medical Review/Hospital Admission is indicated for:**
        - **Dehydration:** Signs include reduced urine output (fewer wet nappies), lethargy, dry mouth, and sunken eyes. If a child is not drinking despite adequate pain relief, they need assessment.
        - **Severe Symptoms or Complications:** While rare, HFMD (particularly from Enterovirus 71) can lead to serious neurological (encephalitis, aseptic meningitis) or cardiopulmonary (myocarditis, pulmonary oedema) complications. Red flags include high fever lasting more than 72 hours,

abnormal or jerking movements, rapid breathing, drowsiness, paleness, or seizures. Any of these symptoms warrant immediate hospital evaluation.

- **Patient Education and Public Health**
  - **Transmission:** Parents should be educated that the virus is highly contagious and spreads easily through saliva, blister fluid, and faeces. It is important to note that viral shedding in faeces can persist for several weeks after recovery.
  - **Hygiene Measures:** Emphasize meticulous handwashing with soap and water, especially after changing nappies and before preparing food. Contaminated surfaces and toys should be cleaned thoroughly. Sharing of cups, cutlery, and towels should be avoided.
  - **Exclusion from Childcare/School:** To prevent outbreaks, children with HFMD must be excluded from childcare, kindergarten, or school. They can return once all the blisters have dried up completely.

Roseola Infantum (Exanthem Subitum)

- **Pathophysiology and Epidemiology**
  - Roseola infantum, also known as 'sixth disease', is a benign viral illness caused by Human Herpesvirus 6 (HHV-6) and occasionally Human Herpesvirus 7 (HHV-7).
  - It is extremely common, affecting most children between the ages of six months and two years. It is estimated that 95% of children have been infected by the age of two.
- **Clinical Presentation and Key Diagnostic Features**
  - **Classic Biphasic Pattern:** The clinical course is highly characteristic. The illness begins with a sudden onset of high fever, often reaching 40^{\circ}C, which lasts for three to five days. A key feature is that despite the high fever, the child often appears remarkably well and remains active.
  - **Rash Appearance:** The hallmark of roseola is the appearance of a rash *after* the fever abruptly subsides (defervescence). The rash is a blanching, rose-pink maculopapular eruption.
  - **Distribution:** The rash typically starts on the trunk and then spreads to the neck and extremities. The face is often spared, which is a useful distinguishing feature from measles. The rash is non-pruritic and fades within two days.
  - **Associated Signs:** Other signs may include cervical lymphadenopathy and Nagayama spots (erythematous papules on the soft palate and uvula), although the latter are not commonly looked for.
- **Comprehensive Management Plan**
  - **Supportive Care:** Treatment is supportive. The main goals are to manage the fever for the child's comfort and to ensure adequate fluid intake to prevent dehydration.
  - **Antipyretics:** Paracetamol or ibuprofen can be used to manage the fever if the child is uncomfortable or irritable.
  - **Febrile Convulsions:** The rapid rise in temperature associated with roseola can be a trigger for a simple febrile convulsion. While frightening for parents, these are typically benign and self-limiting. Management involves standard seizure first aid (placing the child in the recovery position, not restraining them, and not putting anything in their mouth). A seizure lasting longer than five minutes is a medical emergency requiring an ambulance.
- **Referral Pathways and Red Flags**
  - Medical review is warranted if the child is lethargic, shows signs of dehydration, or if

the fever persists for more than 48 hours without an identifiable cause.
    - As with any febrile rash, the presence of non-blanching spots is a red flag for meningococcal disease and requires immediate emergency assessment.
- **Patient Education and Public Health**
    - **Reassurance:** It is vital to reassure parents that roseola is a very common, mild, and self-limiting illness that almost never requires medical treatment.
    - **Contagiousness and Exclusion:** A crucial and often misunderstood point is that the virus is contagious *before* the symptoms (fever and rash) appear. Once the rash is visible, the child is no longer considered infectious. Therefore, if the child is feeling well, they do not need to be excluded from childcare or school. This differs significantly from the exclusion advice for HFMD or chickenpox.

Gianotti-Crosti Syndrome (Papular Acrodermatitis of Childhood)

- **Pathophysiology and Epidemiology**
    - Gianotti-Crosti syndrome (GCS) is not a direct viral infection but rather a distinctive cutaneous immune reaction pattern to a preceding infection or, less commonly, a vaccination.
    - A wide range of viral triggers have been identified, including Epstein-Barr virus (EBV), cytomegalovirus, Coxsackievirus, and parvovirus B19.
    - Historically, GCS was strongly associated with Hepatitis B virus infection. In a highly vaccinated population like Australia, EBV is now the most common trigger. However, the association with Hepatitis B remains clinically important, especially in unimmunised children or those from endemic regions.
    - The condition typically affects children between the ages of one and six years.
- **Clinical Presentation and Key Diagnostic Features**
    - **Prodrome:** There is often a history of a preceding illness with fever and upper respiratory tract symptoms a week or two before the rash appears.
    - **Rash Characteristics:** The eruption consists of monomorphic (all lesions look similar), flat-topped, skin-coloured to pink-brown papules or papulovesicles, typically 1-10 mm in diameter.
    - **Classic Acral Distribution:** The rash is symmetrically distributed on the extensor surfaces of the limbs, the buttocks, and the face. A key diagnostic feature is the relative or complete sparing of the trunk.
    - **Duration and Symptoms:** The rash is long-lasting, persisting for at least 10 days and often for three to eight weeks before spontaneously resolving. It is usually asymptomatic or only mildly itchy; significant pruritus should prompt consideration of other diagnoses like scabies.
- **Comprehensive Management Plan**
    - **Supportive Care and Reassurance:** GCS is a benign and self-limiting condition. The cornerstone of management is reassuring the parents that despite its dramatic appearance and long duration, the rash will resolve completely without scarring.
    - **Symptomatic Treatment:** If pruritus is present, simple emollients or a short course of oral antihistamines may provide some relief. Topical corticosteroids are generally not effective.
    - **Investigations:** In most cases, no investigations are required. However, demonstrating an understanding of the historical and ongoing relevance of the Hepatitis B association is crucial in an exam setting. In an unimmunised child or a

child with risk factors for Hepatitis B exposure, it is appropriate to perform liver function tests and Hepatitis B serology.

- **Patient Education and Public Health**
    - The primary role of the clinician is to provide a confident diagnosis and reassure the family about the benign nature of the condition.
    - Parents should be informed that the rash can last for many weeks but will eventually resolve on its own.
    - Once the rash has appeared, the child is no longer contagious from the initial triggering infection and does not need to be excluded from childcare or school.

Common Childhood Exanthems: A Comparative Overview

To aid in the rapid differentiation of common childhood rashes, a core skill for both clinical practice and examinations, the following table summarises the key features of the most prevalent viral exanthems seen in Australia.

Feature	Measles (Rubeola)	Rubella (German Measles)	Roseola Infantum (Exanthem Subitum)	Erythema Infectiosum (Fifth Disease)	Hand, Foot, and Mouth Disease (HFMD)
Causative Agent	Measles virus (Paramyxovirus)	Rubella virus (Togavirus)	Human Herpesvirus 6 (HHV-6) & 7	Parvovirus B19	Enteroviruses (e.g., Coxsackievirus A16)
Prodrome	3-4 days of high fever, cough, coryza, conjunctivitis (the '3 Cs'). Child appears very unwell.	Mild fever, malaise, prominent posterior auricular and suboccipital lymphadenopathy.	3-5 days of high fever (>39^{\circ}C); child often appears well despite fever.	Mild prodrome of low-grade fever, headache, malaise. Often asymptomatic.	1-2 days of low-grade fever, sore throat, malaise.
Key Oral Signs	**Koplik's spots**: Grey-white papules on buccal mucosa opposite molars. Pathognomonic.	**Forchheimer spots**: Petechiae on soft palate (non-specific).	**Nagayama spots**: Erythematous papules on soft palate/uvula (uncommon).	None specific.	Painful oral vesicles and ulcers on tongue and buccal mucosa.
Rash Characteristics	Maculopapular, erythematous, becomes confluent. Non-pruritic.	Pink maculopapular rash, not confluent.	Rose-pink maculopapular rash. Blanching.	**Stage 1**: "Slapped cheeks" - bright red confluent erythema on face. **Stage 2**: Lacy, reticular	Grey, elliptical vesicles on an erythematous base.

Feature	Measles (Rubeola)	Rubella (German Measles)	Roseola Infantum (Exanthem Subitum)	Erythema Infectiosum (Fifth Disease)	Hand, Foot, and Mouth Disease (HFMD)
				erythema on limbs.	
Rash Distribution	Starts on the face/hairline, spreads downwards (cephalocaudal) to trunk and limbs.	Starts on face, spreads downwards. Fades in the same order.	Starts on the trunk, spreads to neck and extremities. Usually spares the face.	Cheeks first, then a reticular pattern on the trunk and extremities.	**Classic acral distribution**: Palms, soles, and mouth. Often also on buttocks.
Timing of Rash	Appears after the prodrome, concurrently with ongoing fever.	Appears with mild systemic symptoms.	Appears *after* the fever has resolved abruptly.	Appears after the mild prodromal phase.	Appears with or shortly after the onset of fever and oral lesions.
Key Clinical Pearls & Complications	Highly contagious. Notifiable disease. Complications: otitis media, pneumonia, encephalitis, SSPE.	Teratogenic: congenital rubella syndrome (cataracts, deafness, cardiac defects) if infection occurs in first trimester.	Can cause febrile convulsions due to rapid temperature rise.	Can cause aplastic crisis in patients with haemolytic anaemias. Risk of hydrops fetalis in pregnancy.	Highly contagious. Neurological/cardiac complications are rare but serious (associated with EV71).
Exclusion from School/Childcare	Exclude for at least 4 days after the onset of the rash.	Exclude for 4 days after the onset of the rash.	No exclusion required once the rash has appeared and the child is well.	Not required once the rash appears (no longer infectious).	Exclude until all blisters have dried up.

Section 2: Infestations and Allergic Reactions

Scabies

- **Pathophysiology and Epidemiology**
  - Scabies is a parasitic infestation of the skin caused by the mite *Sarcoptes scabiei var. hominis*. The mite burrows into the stratum corneum to lay its eggs.
  - It is transmitted through direct and prolonged skin-to-skin contact, making it common within households and institutions. It is frequently sexually acquired in adults.
  - The intense itching (pruritus) is not caused by the burrowing itself, but by a delayed type-IV hypersensitivity reaction to the mite, its eggs, and its faeces. This explains why symptoms can take 3-6 weeks to appear after a primary infestation but only 24

hours upon re-infestation.

- **Clinical Presentation and Key Diagnostic Features**
  - **Cardinal Symptom:** The hallmark is severe, generalised pruritus that is characteristically worse at night or after a hot bath or shower.
  - **Lesions:** The rash consists of small, erythematous papules, vesicles, and nodules. Excoriations from intense scratching are almost always present and can lead to secondary bacterial infection.
  - **Pathognomonic Sign (Burrows):** These appear as thin, wavy, greyish or skin-coloured lines on the skin surface. They are most commonly found in the interdigital web spaces of the hands, the flexor surfaces of the wrists, elbows, axillae, belt line, and on the penis and scrotum in men and around the nipples in women. The head and neck are characteristically spared in adults and older children.
  - **Diagnosis:** Diagnosis is often clinical, based on the history of intense nocturnal itch and the characteristic distribution of lesions. The presence of burrows is diagnostic. Dermatoscopy can be used to visualise the mite at the end of a burrow (the 'delta wing' or 'jet with contrail' sign). Definitive diagnosis can be made by microscopic identification of the mite, eggs, or faecal pellets from a skin scraping of a burrow.
- **Comprehensive Management Plan**
  - The core principle of successful scabies management is to treat the infected individual, all household members, and any close personal or sexual contacts simultaneously, and to decontaminate the environment. Failure to adhere to this comprehensive approach is the most common reason for treatment failure and re-infestation.
  - **First-Line Pharmacotherapy (Topical):** In Australia, the treatment of choice is **Permethrin 5% cream**.
    - **Application Instructions:** The cream must be applied to the *entire body surface* from the neck down to the soles of the feet. It is crucial to apply it thoroughly to all skin folds, between fingers and toes, and under the fingernails and toenails. In infants, the elderly, and people living in central and northern Australia, the cream should also be applied to the face, scalp, and neck, avoiding the areas around the eyes and mouth.
    - **Duration:** The cream should be left on for 8 to 14 hours (typically overnight) and then washed off thoroughly.
    - **Repeat Treatment:** The treatment must be repeated exactly 7 days later to kill any mites that have hatched from eggs that survived the first application.
  - **Second-Line Pharmacotherapy (Oral): Oral ivermectin** (200 mcg/kg) is an effective alternative, available on prescription. It is particularly useful for crusted (Norwegian) scabies, in institutional outbreaks where mass treatment is required, or for individuals who cannot comply with topical therapy. It is given as a single dose, which should also be repeated after 7-14 days. It is contraindicated in children weighing less than 15 kg and in pregnant or breastfeeding women.
  - **Management of Contacts:** All household members and close contacts (including sexual partners) from the preceding month must be treated at the same time as the index case, even if they do not have symptoms.
- **Referral Pathways and Red Flags**
  - Referral to a dermatologist or infectious diseases specialist should be considered for cases of treatment failure, atypical presentations, or crusted scabies. Crusted

scabies is a hyper-infestation seen in immunocompromised, elderly, or debilitated individuals and is extremely contagious.
  - Secondary bacterial infection (impetigo, cellulitis) is a common complication due to scratching. If signs of infection (increased redness, pus, crusting) are present, a course of oral antibiotics may be required.
- **Patient Education and Public Health**
  - **Environmental Decontamination:** On the day after the first application of treatment, all bedding, towels, and clothing used by the patient and contacts in the preceding 72 hours should be decontaminated. This is achieved by washing in hot water (>50-60^{\circ}C) and drying in a hot dryer, or by sealing items in a plastic bag for at least 72 hours (some guidelines recommend up to 8 days). Mattresses and furniture should be vacuumed.
  - **Post-Scabetic Itch:** It is essential to warn patients that itching can persist for two to four weeks after successful treatment. This is due to the ongoing hypersensitivity reaction to dead mites and their products and does not signify treatment failure. This itch can be managed with emollients, topical corticosteroids, or oral antihistamines.
  - **Exclusion:** The individual is no longer infectious 24 hours after the first effective treatment and can return to school or work at that time.

Urticaria (Hives)

- **Pathophysiology and Epidemiology**
  - Urticaria is a common skin condition characterized by the development of wheals (hives), angioedema, or both. It affects up to 20% of the population at some point in their lives.
  - It is caused by the degranulation of cutaneous mast cells, which release histamine and other inflammatory mediators, leading to localized vasodilation and increased vascular permeability.
  - **Acute Urticaria** is defined as lasting less than six weeks. It often has an identifiable trigger, such as a viral infection (most common in children), an allergic reaction to food or medication, or an insect bite.
  - **Chronic Urticaria** lasts for more than six weeks. It is further divided into Chronic Spontaneous Urticaria (CSU), where no trigger is found and an autoimmune basis is often suspected, and Chronic Inducible Urticaria, where wheals are triggered by physical stimuli like cold, pressure, or scratching (dermographism).
- **Clinical Presentation and Key Diagnostic Features**
  - **Wheals:** These are the hallmark lesions. They are circumscribed, raised, erythematous plaques with central pallor. They are intensely pruritic and characteristically transient, with individual lesions lasting less than 24 hours before disappearing and reappearing elsewhere.
  - **Angioedema:** This is a deeper, less well-defined swelling of the dermis and subcutaneous tissues. It commonly affects areas with lax tissue, such as the eyelids, lips, and genitalia. It is often described as painful or burning rather than itchy and resolves more slowly than wheals (up to 72 hours).
  - **Diagnosis:** The diagnosis is clinical, based on the characteristic appearance of the lesions and a thorough history. A detailed history is crucial to identify potential triggers for acute urticaria or patterns in chronic inducible urticaria. Routine allergy

testing is not indicated for chronic spontaneous urticaria as an allergic cause is rare.

- **Comprehensive Management Plan**
  - **Acute Urticaria:**
    - **First-Line Treatment:** Modern, second-generation, non-sedating H1-receptor antihistamines (e.g., cetirizine, loratadine, fexofenadine) are the mainstay of treatment. They should be taken regularly to suppress symptoms.
    - **Severe Cases:** For severe, widespread urticaria that is not responding to antihistamines, a short course of oral prednisolone (e.g., 0.5–1 mg/kg, max 60 mg, for 3–5 days) may be considered to gain control.
  - **Chronic Spontaneous Urticaria (CSU):** Management follows a stepwise approach based on Australian guidelines.
    - **Step 1:** Regular daily use of a licensed dose of a second-generation non-sedating antihistamine.
    - **Step 2:** If symptoms are not controlled after 2-4 weeks, the dose of the non-sedating antihistamine can be increased, up to four times the standard licensed dose, under medical supervision.
    - **Step 3:** If symptoms persist, the patient should be referred to a clinical immunologist/allergist or a dermatologist for further management.
    - **Step 4 (Specialist Treatment):** For patients with severe CSU refractory to high-dose antihistamines, the addition of **omalizumab** (an anti-IgE monoclonal antibody) is an effective option. Omalizumab is subsidised on the Pharmaceutical Benefits Scheme (PBS) in Australia for this indication when prescribed by an appropriate specialist. Other specialist options include immunosuppressants like cyclosporin.
- **Referral Pathways and Red Flags**
  - **Anaphylaxis:** The most important red flag is the presence of systemic symptoms along with urticaria or angioedema. Any involvement of the respiratory system (difficult/noisy breathing, wheeze, persistent cough, hoarse voice), cardiovascular system (persistent dizziness, collapse), or severe gastrointestinal symptoms indicates anaphylaxis. This is a medical emergency requiring immediate administration of intramuscular adrenaline (epinephrine) and transfer to hospital.
  - **Urticarial Vasculitis:** This should be suspected if individual wheals are painful, persist for more than 24-48 hours, and leave behind bruising or pigmentation. This requires a skin biopsy for diagnosis and investigation for underlying systemic autoimmune conditions.
- **Patient Education**
  - **Trigger Avoidance:** Advise patients to avoid any identified triggers. Common non-specific aggravating factors for chronic urticaria include aspirin and other NSAIDs, alcohol, excessive heat, and stress.
  - **Chronic Urticaria:** Explain that CSU is often a long-term condition that can last for months or years, and the goal of treatment is to control symptoms and improve quality of life. Reassure them that it is not contagious and, in the absence of systemic symptoms, not life-threatening.
  - **Antihistamine Use:** Emphasize that for chronic urticaria, antihistamines are most effective when taken daily to prevent wheals from forming, rather than being used "as needed" once symptoms are severe.

Section 3: Bacterial Skin Infections

Staphylococcus aureus Skin Infections

- **Pathophysiology and Epidemiology**
  - *Staphylococcus aureus* is a Gram-positive bacterium that is a common commensal of the skin and anterior nares in up to 30% of the healthy population.
  - Infection occurs when the bacterium breaches the skin barrier, for example, through a cut, graze, or in areas of pre-existing skin disease like eczema.
  - Methicillin-Resistant *Staphylococcus aureus* (MRSA) is a significant public health concern in Australia. It is important to distinguish between community-associated MRSA (CA-MRSA), which can cause infections in healthy individuals, and hospital-associated MRSA (HA-MRSA). CA-MRSA often retains susceptibility to non-beta-lactam antibiotics, whereas HA-MRSA is frequently multi-resistant.
- **Clinical Presentations**
  - **Impetigo (School Sores):** A highly contagious, superficial skin infection common in children. It presents as vesicles or pustules that rupture to form characteristic honey-coloured crusts, typically around the mouth and nose.
  - **Folliculitis, Furuncles (Boils), and Carbuncles:** These represent a spectrum of infection of the hair follicle, from superficial pustules (folliculitis) to deep, painful, pus-filled nodules (furuncles), and coalescing furuncles (carbuncles).
  - **Cellulitis:** An infection of the deeper dermis and subcutaneous tissue. It presents as a poorly demarcated, expanding area of erythema, warmth, swelling, and tenderness. A portal of entry may be visible. Erysipelas, a key differential, is more superficial, has a sharply demarcated, raised border, and is more commonly caused by *Streptococcus pyogenes*.
  - **Staphylococcal Scalded Skin Syndrome (SSSS):** A severe, toxin-mediated disease primarily affecting children under five. It is caused by exfoliative toxins that lead to widespread erythema and superficial blistering and peeling of the skin, giving it a scalded appearance. This is a medical emergency requiring hospitalisation.
- **Comprehensive Management Plan (Australian Context)**
  - The choice of antibiotic therapy in Australia is critically dependent on the clinical severity, the likely pathogen (*Staphylococcus* vs. *Streptococcus*), and the local prevalence of MRSA. The Australian Therapeutic Guidelines are the definitive resource for prescribing.
  - **Incision and Drainage:** For any localised collection of pus (abscess, furuncle, carbuncle), incision and drainage is the primary and often sufficient treatment. A swab of the pus should be sent for microscopy, culture, and sensitivities.
  - **Antibiotic Selection (based on Therapeutic Guidelines):**
    - **Mild to Moderate Non-purulent Cellulitis or Impetigo (likely MSSA or *Streptococcus*):** First-line treatment is with an oral anti-staphylococcal beta-lactam, such as **flucloxacillin**, **dicloxacillin**, or **cefalexin**.
    - **Suspected CA-MRSA (e.g., recurrent boils/abscesses, specific risk factors):** If an oral antibiotic is required after drainage, choices should be guided by sensitivities if available. Empiric options for non-severe infections

include **trimethoprim/sulfamethoxazole**, **clindamycin**, or **doxycycline**.
        - **Severe Infections (requiring hospitalisation):** Intravenous antibiotics are required. For severe cellulitis likely due to MSSA/Strep, **IV flucloxacillin** or **cefazolin** is used. If MRSA is suspected or confirmed, **IV vancomycin** is the standard of care.
    - **Recurrent MRSA Infections - Decolonisation Protocol:** This is a key strategy for managing recurrent staphylococcal skin infections, particularly MRSA. The protocol should only be initiated after the acute infection has been successfully treated.
        - **Regimen (typically for 5-7 days):**
            1. **Nasal Decolonisation:** Apply **mupirocin 2% nasal ointment** to the inside of both nostrils twice daily.
            2. **Skin Decolonisation:** Use a daily antiseptic body wash, such as **2% chlorhexidine**, concentrating on axillae and the perineum. An alternative is a daily bleach bath (e.g., 60mL of 6% household bleach in a standard-sized bath for 15 minutes).
        - **Household Management:** If infections are ongoing within a household, simultaneous decolonisation of all members should be considered. Environmental hygiene, including hot-washing of all linen, towels, and clothing, is also crucial.
- **Patient Education and Public Health**
    - **Hygiene:** Meticulous personal hygiene is paramount. This includes frequent handwashing with soap and water, keeping wounds clean and covered with waterproof dressings, and avoiding the sharing of personal items like towels, razors, and clothing.
    - **Exclusion:** Children with impetigo should be excluded from school or childcare until they have received at least 24 hours of antibiotic treatment and their sores are covered with a watertight dressing.
    - **Antibiotic Stewardship:** Educate patients on the importance of completing the full course of prescribed antibiotics to prevent the development of antibiotic resistance.

Section 4: Complex and Congenital Dermatoses

Pyoderma Gangrenosum (PG)

- **Pathophysiology and Epidemiology**
    - Pyoderma gangrenosum (PG) is a rare, non-infectious, ulcerative neutrophilic dermatosis. It is a manifestation of an autoinflammatory process, not a primary infection or a form of gangrene.
    - In over 50% of cases, PG is associated with an underlying systemic disease. The most common associations in Australia and worldwide are inflammatory bowel disease (IBD - Crohn's disease and ulcerative colitis), inflammatory arthritis (e.g., rheumatoid arthritis), and haematological disorders (e.g., myelodysplastic syndrome, monoclonal gammopathy).
- **Clinical Presentation and Key Diagnostic Features**
    - **Classic Ulcer:** The typical lesion begins as a tender, inflammatory papule, pustule, or nodule that rapidly breaks down to form a progressively enlarging, exquisitely painful ulcer.

    - **Pathognomonic Border:** The ulcer has a characteristic violaceous (purple-red), irregular, and undermined border, where the edge of the ulcer appears to be lifting away from the base.
    - **Pathergy:** A key clinical feature is pathergy, where new lesions develop or existing ulcers worsen at sites of minor trauma. This is a critical concept, as inappropriate surgical debridement can lead to catastrophic enlargement of the ulcer.
    - **Diagnosis:** PG is a diagnosis of exclusion. There are no specific serological or histological markers. The diagnosis is made based on the recognition of the characteristic clinical features after meticulously ruling out other causes of cutaneous ulceration, such as infection, vasculitis, malignancy, and venous or arterial insufficiency. A deep incisional or punch biopsy from the active, violaceous border is essential, not to confirm PG, but to exclude these mimics. The histology in PG is non-specific, showing a dense neutrophilic infiltrate.
- **Comprehensive Management Plan**
    - The management of PG is complex and requires a multidisciplinary approach, often led by a dermatologist. The primary goals are to control the inflammation, manage pain, and promote wound healing.
    - **Avoid Aggressive Debridement:** The most crucial initial step is to avoid aggressive surgical debridement until a diagnosis is established and systemic immunosuppressive therapy has been initiated. Debridement can trigger pathergy and cause rapid ulcer expansion.
    - **Local Wound Care:** Gentle wound care with non-adherent dressings is important. Once the inflammation is controlled and arterial disease is excluded, compression therapy can aid healing of lower leg ulcers.
    - **Topical Therapy (for mild or superficial disease):** High-potency topical corticosteroids or topical tacrolimus can be applied to the ulcer and its border. Intralesional corticosteroid injections into the active border may also be effective.
    - **Systemic Therapy (First-Line):** For rapidly progressing or large ulcers, systemic therapy is required. High-dose oral **prednisolone** (e.g., 0.5–1 mg/kg/day) is the mainstay of treatment to achieve rapid control of inflammation.
    - **Steroid-Sparing Agents:** Due to the long-term side effects of corticosteroids, a steroid-sparing agent is often introduced early. **Ciclosporin** is a common choice and has evidence comparable to prednisolone. Other options include mycophenolate mofetil, azathioprine, and dapsone.
    - **Biologic Agents:** For severe or refractory disease, TNF-alpha inhibitors such as **infliximab** or **adalimumab** have shown significant efficacy. In Australia, their use for PG is considered 'off-label' and is typically managed by a specialist.
    - **Intravenous Immunoglobulin (IVIg):** IVIg is reserved for severe, refractory cases that have failed other treatments. Its use in Australia is governed by specific criteria under the National Blood Authority.
- **Referral Pathways**
    - Any patient with a suspected diagnosis of PG requires urgent referral to a dermatologist for confirmation and management.
    - A multidisciplinary team approach is essential, involving specialists from gastroenterology, rheumatology, or haematology to investigate and manage any underlying systemic disease.

Port-Wine Syndrome (Capillary Malformation)

- **Pathophysiology and Epidemiology**
  - A port-wine stain (PWS), more accurately termed a capillary malformation, is a congenital, low-flow vascular malformation involving post-capillary venules in the dermis.
  - It is caused by a somatic mutation in the *GNAQ* gene that occurs during embryogenesis. It is not inherited.
  - A PWS is present at birth and persists throughout life. It grows in proportion to the child and does not regress spontaneously.
- **Clinical Presentation and Key Diagnostic Features**
  - **Appearance:** At birth, it presents as a flat, well-demarcated, pink or red patch of skin.
  - **Evolution:** Over time, the lesion tends to darken to a deep red or purple colour. The skin may also become thickened, and vascular nodules or blebs can develop on the surface, which may bleed.
  - **Location:** Can occur anywhere on the body but is most common on the face and neck. Diagnosis is clinical.
- **Comprehensive Management Plan**
  - **Laser Therapy:** The gold standard and most effective treatment is the **pulsed dye laser (PDL)**, such as the V Beam laser. The laser selectively targets haemoglobin within the dilated blood vessels, causing thermal damage and vessel closure with minimal damage to the surrounding skin.
    - **Treatment Course:** Multiple treatment sessions (often 10 or more) are required, typically spaced 4-8 weeks apart. Complete clearance is achieved in a minority of patients (10-20%), but significant lightening (>50%) is common.
    - **Timing of Treatment:** Treatment can be initiated in infancy. Starting early may lead to better outcomes and can sometimes be performed in very young infants without the need for general anaesthesia.
  - **General Measures:** Lifelong, diligent sun protection is important as UV exposure can cause the PWS to darken. Regular use of moisturiser can prevent dryness over the affected skin.
  - **Cosmetic Camouflage:** Specialised, opaque makeup can be used to effectively cover the PWS for cosmetic purposes.
- **Associated Syndromes and Red Flags**
  - A PWS can be a cutaneous marker for more complex syndromes. A high index of suspicion is required for certain locations.
  - **Sturge-Weber Syndrome:** This is the most critical association. A PWS involving the ophthalmic (V1) distribution of the trigeminal nerve (forehead and upper eyelid) raises suspicion for this syndrome. It is associated with ipsilateral leptomeningeal angiomas, which can cause seizures, developmental delay, and hemiparesis, as well as glaucoma on the affected side. Any infant with a V1 PWS requires urgent referral to a paediatric ophthalmologist for glaucoma screening and a paediatric neurologist for assessment and consideration of brain imaging (MRI).
  - **Klippel-Trenaunay Syndrome:** A PWS affecting a limb may be associated with underlying venous and lymphatic malformations, as well as bony and soft tissue

hypertrophy of that limb. These patients require referral to a multidisciplinary vascular anomalies clinic.

- **Patient Education**
    - It is essential to explain that a PWS is a permanent birthmark that will not resolve on its own.
    - Parents should be counselled on the goals, expectations, and limitations of laser therapy. It is a process of lightening and management, not necessarily a "cure".
    - Provide information and support regarding associated syndromes if the location of the PWS is concerning.

Epidermolysis Bullosa (EB)

- **Pathophysiology and Epidemiology**
    - Epidermolysis Bullosa (EB) is a group of rare, inherited genetic disorders characterized by extreme fragility of the skin and mucous membranes, where minor mechanical friction or trauma leads to blistering.
    - It is caused by mutations in genes that code for structural proteins essential for adhesion between the epidermal and dermal layers of the skin. There are four main types (Simplex, Junctional, Dystrophic, and Kindler syndrome), with numerous subtypes, each varying in severity and inheritance pattern (autosomal dominant or recessive).
- **Clinical Presentation and Key Diagnostic Features**
    - **Hallmark:** The formation of blisters and erosions in response to minimal trauma. This is typically present from birth or within the first few weeks of life.
    - **Spectrum of Severity:** The clinical presentation varies widely.
        - **Mild forms (e.g., localized EB Simplex):** May only have blistering on the hands and feet.
        - **Severe forms (e.g., severe Junctional or Dystrophic EB):** Can involve widespread, debilitating blistering, extensive scarring, fusion of the fingers and toes (pseudosyndactyly or "mitten" deformities), and significant extracutaneous involvement.
    - **Extracutaneous Manifestations:** Blistering can affect mucosal surfaces, including the oral cavity, oesophagus, and gastrointestinal tract, leading to painful eating, dysphagia, oesophageal strictures, malnutrition, and chronic constipation. Dental, nail, and eye problems are also common.
    - **Diagnosis:** Suspected clinically in a neonate with blistering. Definitive diagnosis requires a skin biopsy for specialised analysis (immunofluorescence antigen mapping and transmission electron microscopy) to identify the level of skin cleavage, followed by genetic testing to confirm the specific gene mutation.
- **Comprehensive Management Plan**
    - Management of EB is highly complex and requires a coordinated, lifelong multidisciplinary team approach. While the primary manifestations are cutaneous, the management priorities are often systemic, focusing on wound care, pain control, nutritional support, and management of complications.
    - **Wound Care:** This is a cornerstone of daily management.
        - **Blister Management:** New blisters should be lanced with a sterile needle to drain fluid and prevent enlargement, while leaving the blister roof intact to act as a natural dressing.

        - **Dressings:** Non-adherent, protective dressings are used to cover wounds, prevent further trauma, and manage exudate.
        - **National Epidermolysis Bullosa Dressing Scheme (NEBDS):** This is a vital, federally-funded Australian program that provides subsidised access to specialised dressings for eligible individuals with EB. Awareness of this scheme is essential for practicing in Australia.
    - **Infection Prevention and Management:** Wounds are highly susceptible to infection. Regular antiseptic washes (e.g., bleach baths) and judicious use of topical and systemic antibiotics are required to manage wound infections.
    - **Pain Management:** Pain is a significant and constant issue, especially during dressing changes. A structured pain management plan, often developed with a paediatric pain service, is essential and may involve a combination of simple analgesics, opioids, and non-pharmacological techniques.
    - **Nutritional Support:** Patients with EB, particularly severe forms, have increased metabolic demands due to chronic wounding and inflammation. A high-calorie, high-protein diet is necessary to support healing and growth. A dietitian is a core member of the multidisciplinary team. Oesophageal involvement may necessitate soft diets or gastrostomy tube feeding.
    - **Management of Complications:**
        - **Physiotherapy and Occupational Therapy:** To prevent and manage contractures and maintain function, especially of the hands.
        - **Dental Care:** Requires specialist dental care due to enamel defects and oral blistering.
        - **Cancer Surveillance:** Patients with severe Dystrophic EB have a very high lifetime risk of developing aggressive squamous cell carcinomas in areas of chronic wounding. Regular skin surveillance is a critical part of long-term care.
- **Referral Pathways**
    - Any neonate or child with suspected EB requires immediate referral to a tertiary paediatric centre with a specialised, multidisciplinary EB service.
    - Genetic counselling must be offered to the family to discuss the inheritance pattern, prognosis, and options for future pregnancies.

Conclusions

The effective management of these diverse dermatological conditions in the Australian context requires a strong foundation in clinical diagnosis, a nuanced understanding of local treatment guidelines, and a commitment to patient-centred education. For the common viral exanthems of childhood, the clinician's primary role is accurate differentiation, provision of supportive care advice, and vigilant identification of red flags for serious secondary complications or mimics like meningococcal disease. In managing conditions like scabies and *S. aureus* infections, success hinges not only on correct pharmacotherapy but also on implementing robust public health measures, such as simultaneous contact treatment and environmental decontamination for scabies, and targeted decolonisation protocols for recurrent MRSA.

For the more complex and chronic dermatoses, such as pyoderma gangrenosum, port-wine syndrome, and epidermolysis bullosa, a multidisciplinary approach is paramount. The clinician must recognise the systemic associations of these conditions and facilitate timely referral to specialist services. A critical aspect of practice in Australia involves awareness of and navigation through specific national programs, such as the National Epidermolysis Bullosa Dressing Scheme and PBS criteria for specialist medications like omalizumab. Mastering these clinical nuances—from the bedside press test for a febrile rash to counselling a family on a comprehensive MRSA decolonisation plan—is the hallmark of a competent and safe practitioner prepared for both the AMC CAT examination and the demands of Australian clinical practice.

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